

Multilevel Methods on Finite Elements

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Modeling Physics

- ▶ Problems include modeling how water flows from a source through porous materials.
- ▶ Heat dissipating through a material.

get graphics from that poisson physics youtube

Instead of words, I would put the some diff eqs and pdes here and we can verbally say what they are. May be easier to convey the difficulty/complexity of modeling a continuous thing that does not have a simple closed form solution. Could also pull small videos or gifs from online to show the complexity (like your comment)

PDEs and Finite Elements

- 1 We model these systems with the Poisson equation
- 2 Then we use finite elements to approximate the solution
- 3 Linear algebra is used in making those approximations

from the books

Maybe an image of something from COMSOL that has an FEA grid for illustration? I liked the blade here <https://www.comsol.com/multiphysics/fea-software>. or if there is a good picture in Panayots book?

PDE

keep it brief

- 1 Start with a PDE
- 2 Weak form
- 3 Divergence theorem/integrate by parts
- 4 Discretize the domain
- 5 Represent the weak form as a discretization on the parts of the mesh
- 6 The operators become matrices
- 7 Solve with linear algebra
- 8 Use norms to see how good your approximation is to toy problems where you know the solution

I thought these were good at explaining some of the concepts

- stiffness matrix walk through: https://www.ccg.msm.cam.ac.uk/system/files/documents/FEMOR_Lecture_1.pdf

- bases visualization: <https://www.comsol.com/multiphysics/finite-element-method>

- fun basis viz: <https://fea.ju.se/weakForm1D/>

Really Big Problems

Too big to run

Like Hillary said yesterday, some of these problems are absurdly large. We have to find ways to solve them when $x = A^{(-1)}b$

formatting issue, probably need to wrap in $\{ \}$

In the spirit of the class we will use Gauss Siedel smoothing and a multilevel method with a coarse direct solve and use that as the preconditioner for preconditioned conjugate gradient.

NGSolve

Mention NGSolve, what it is and does

A Hierarchy of Finite Element Spaces

We built this cool structure

Here's what it is like

it's a bunch of levels of finite element spaces with nested mesh sizes

Setting up the spaces

At each level, we have our pde representation that we need to solve
and we need to use P matrices to resize things for sending between levels
and we want to solve things or smooth them
and measure norms

Multilevel Interactions

once we have built the P 's we can solve our problem with a v-cycle
show v-cycle pic

Solving V-cycle with smoothing and coarse solve

this was our first goal we got it working, eventually see how it works!

Then we compared different sizes

for different sizes of spaces, you get different results in CG but with this method, we get size independent results

Now we use it as a preconditioner

We made it so that we run PCG and when it asks for a preconditioner, it gets our two-level method See how it works

Results and Plots

not sure how many plots we have to go here

Conclusion

- ▶ We did this
- ▶ We didn't do that
- ▶ This is what worked
- ▶ This is what we would do next